



1. Data Cleaning

The training set contained 10,050 rows and 25 columns. No missing values were present in the test set.

1.1 Missing Values

Two tiers of missing data were identified. Ten rows were each missing approximately 20 columns including the target and were dropped entirely. The remaining ~100 rows missing one or two values were imputed using median for numeric columns and mode for Status. The imputer was fitted on train only to prevent data leakage. Table 1 summarises the treatment.

Table 1: Missing Value Summary and Treatment

Issue	Rows Affected	% of Dataset	Action Taken
Rows missing >5 columns	10	0.10%	Dropped entirely
Remaining numeric columns	~100	<1%	Median imputation
Status column (categorical)	~20	<0.2%	Mode imputation

1.2 Outlier Detection and Treatment

Outliers were identified using the IQR method [$Q1 - 1.5 \times IQR$, $Q3 + 1.5 \times IQR$]. Tank Failure Pressure had 50 values up to 4,995 bar — far beyond the physically plausible LPG range of 5–60 bar — and was capped at the 99th percentile (~57 bar). Sensor Position coordinates and Vapour Height were flagged but retained as physically valid. The 881 flagged Target Pressure values represent legitimate high-pressure blast readings and were addressed via log-transformation rather than removal. Table 2 summarises decisions.

Table 2: Outlier Detection and Treatment Summary

Column	Outliers	Range	Treatment
Tank Failure Pressure (bar)	50	5–4,996	Capped at 99th percentile (~57 bar)
Sensor Position y, z	102	Valid coords	Retained — physically valid sensor positions
Vapour Height (m)	71	0.18–2.62	Retained — natural right skew, not errors
Target Pressure (bar)	881	0.02–9.17	Log-transformed — real physics, not errors

1.3 Duplicate Removal

Fifty exact duplicate rows were found clustered at rows 9,990–10,039, suggesting an accidental double-append during data export. These were removed using `drop_duplicates(keep='first')`, reducing the training set from 10,040 to 9,990 rows. No duplicates existed in the test set.

1.4 Correcting Inaccurate Entries

The Status column contained nine inconsistent string variants of two categories. These were standardised using a mapping dictionary and binary-encoded (Superheated = 1, Subcooled = 0). Table 3 shows all variants.

Table 3: Status Column Standardisation

Raw Value	Corrected To
Superheated, superheated, Superheat, Saperheated	Superheated
Subcooled, subcooled, Subcool, Subcoled	Subcooled

2. Data Processing

2.1 Feature Selection

Feature selection used Pearson correlation and Random Forest feature importance to identify redundant columns. Four features — Status, Liquid Critical Pressure, Liquid Boiling Temperature, and Liquid Critical Temperature — scored below 0.3% RF importance. These are physical constants for only two substances, making them inferrable from correlated temperature and pressure columns. Cross-validation confirmed dropping them reduced R^2 by only 0.0001. Table 4 summarises selected features.

Table 4: Feature Selection Summary (selected rows shown)

Feature	Corr r	RF Importance	Decision
Sensor Position Side	0.006	32.8%	Retained
Sensor ID	0.065	25.9%	Retained
Vapour Height (m)	0.272	4.9%	Retained
Status	0.026	0.3%	Dropped
Liquid Critical Pressure / Temp / Boiling Temp	~0.045	<0.2%	Dropped

2.2 Feature Engineering

Four physics-based features were derived to reduce the burden on models to learn implicit combinations. Tank Volume (Width × Length × Height) captures total stored energy and ranked 3rd in RF importance at 9.5%. Sensor Distance to BLEVE ($\sqrt{x^2+y^2+z^2}$) summarises 3D distance from the blast source, ranked 4th at 8.5%. Obstacle Volume and Obstacle Face Area capture wall mass and blast interception respectively. Cross-validated R^2 improved from 0.8735 to 0.8876 after adding these features. Table 5 summarises them.

Table 5: Engineered Features Summary

Feature	Formula	RF Importance	Rationale
Tank Volume (m ³)	Width × Length × Height	9.5%	Total stored energy
Sensor Distance to BLEVE (m)	$\sqrt{x^2 + y^2 + z^2}$	8.5%	Pressure decay with distance
Obstacle Volume (m ³)	Width × Height × Thickness	1.5%	Wall mass and reflection
Obstacle Face Area (m ²)	Width × Height	0.9%	Blast interception area

2.3 Data Type Conversion

Status was binary encoded (Superheated = 1, Subcooled = 0). One-hot encoding was unnecessary given exactly two categories. Sensor Position Side contained integer codes already and required no transformation.

2.4 Feature Scaling

Features span very different ranges (e.g. Vapour Temperature: 281–573 K vs Liquid Ratio: 0.1–0.9). StandardScaler ($z = (x - \mu) / \sigma$) was applied for models sensitive to magnitude. It was preferred over MinMaxScaler as it is more robust to the residual skew remaining after outlier treatment. The scaler was fitted on train only and applied to test, preventing leakage. Table 6 shows which models used scaling.

Table 6: Feature Scaling by Model

Model	Scaling	Reason	Feature Set
Ridge Regression	StandardScaler	Coefficients sensitive to magnitude	X_scaled

Model	Scaling	Reason	Feature Set
Neural Network	StandardScaler	Gradient descent diverges on unscaled inputs	X_scaled
Random Forest	None	Tree splits are scale-invariant	X (unscaled)

3. Model Selection

Three fundamentally different model types were selected: Ridge Regression (linear/parametric), Random Forest (tree ensemble), and a Multilayer Perceptron Neural Network (gradient-based). All were first benchmarked at default hyperparameters under 5-fold CV on the log-transformed target, with MAPE back-computed in bar scale.

3.1 Ridge Regression

Ridge adds L2 regularisation to Ordinary Least Squares, selected as a linear baseline. BLEVE blast physics are inherently non-linear, so underfitting was expected. Training and CV R^2 were both ~ 0.73 — confirming consistent underfitting rather than overfitting. Retained for comparison only. Key characteristics: parametric, closed-form, high interpretability, single hyperparameter alpha.

3.2 Random Forest

Random Forest trains an ensemble of decision trees on bootstrapped subsets of data, averaging predictions to reduce variance. It makes no assumptions about the functional form, captures non-linear interactions, and is scale-invariant. At default settings it achieved CV R^2 of 0.8876 and MAPE of 0.7891 — a substantial improvement over Ridge. Key characteristics: non-parametric, tree-based, scale-invariant, medium interpretability.

3.3 Neural Network (MLP)

The MLP learns a continuous differentiable mapping through backpropagation. Unlike RF, it produces a smooth function rather than a piecewise constant approximation, generalising differently between training points. At default settings it achieved the best CV R^2 of 0.9015 and MAPE of 0.6926. Key characteristics: non-parametric, gradient-based, requires scaling, higher risk of overfitting.

3.4 Baseline Comparison and Rejected Models

Table 7 compares default performance. Table 8 summarises model properties. SVR was considered but excluded due to $O(n^2-n^3)$ training complexity on 9,990 rows. XGBoost was excluded as it shares the same tree-based family as Random Forest and would not satisfy the distinct model requirement.

Table 7: Default Hyperparameter Performance (5-fold CV)

Model	R^2	MAPE	RMSE	Observation
Ridge Regression	0.7284	1.4096	0.5161	Underfitting
Random Forest	0.8876	0.7891	0.3202	Strong non-linear capture
Neural Network (MLP)	0.9015	0.6926	0.2981	Best default performance

Table 8: Model Property Comparison

Property	Ridge	Random Forest	Neural Network
Model family	Linear/Parametric	Tree ensemble	Neural network
Handles non-linearity	No	Yes	Yes
Requires scaling	Yes	No	Yes
Training mechanism	Closed-form	Bagged trees	Backpropagation
CV R^2 / MAPE (default)	0.73 / 1.41	0.89 / 0.79	0.90 / 0.69

4. Hyperparameter Tuning

All searches used 5-fold CV scored on R^2 . Strategy differed by search space size.

4.1 Ridge — Grid Search

With only one hyperparameter (alpha), exhaustive Grid Search over six log-spaced values (0.01–1000) was used. Best alpha = 1 ($R^2 = 0.7668$). The flat performance curve confirms the limitation is architectural, not regularisation-related. Table 9 shows full results.

Table 9: Ridge Grid Search Results

Alpha	Mean CV R^2	Mean CV MAPE	Selected
0.01	0.7661	1.4209	No
0.1	0.7666	1.4163	No
1	0.7668	1.4096	Yes ✓
10	0.7665	1.4108	No
100	0.7634	1.4291	No
1000	0.7423	1.5047	No

4.2 Random Forest — Randomised Search (30 iterations)

Five parameters and a large combined search space made Grid Search prohibitive. Randomised Search sampled 30 combinations with random_state=42. Table 10 reports the best configuration. Best CV R^2 improved to 0.8923. Unlimited depth performs best because bagging across 200 trees controls the variance of individual deep trees.

Table 10: Random Forest Best Hyperparameters

Hyperparameter	Best Value	Justification
n_estimators	200	More trees reduce variance without excessive cost
max_depth	None	Full-depth trees — bagging controls overfitting
min_samples_leaf	1	Finest granularity best with ensemble averaging
max_features	'sqrt'	Standard RF setting, best tree diversity

4.3 Neural Network — Randomised Search (20 iterations)

Twenty combinations were evaluated covering architecture, regularisation, and learning rate. Table 11 reports the best configuration. CV R^2 improved to 0.9218. The (100, 50) architecture allows hierarchical feature learning while remaining compact enough to generalise well.

Table 11: Neural Network Best Hyperparameters

Hyperparameter	Best Value	Justification
hidden_layer_sizes	(100, 50)	Two layers allow hierarchical feature learning
alpha	0.001	Moderate regularisation balances fit and generalisation
learning_rate_init	0.001	Standard learning rate — stable convergence

4.4 Tuning Summary

Table 12 compares default and tuned performance. The Neural Network saw the largest gain (MAPE –0.082) and was selected as the final model, retrained on all 9,990 rows before generating test predictions.

Table 12: Default vs Tuned Performance

Model	Default R^2	Tuned R^2	Default MAPE	Tuned MAPE	MAPE Δ	Improvement
Ridge Regression	0.7284	0.7668	1.4096	1.4096	0.000	Minimal
Random Forest	0.8876	0.8923	0.7891	0.7350	-0.054	Moderate
Neural Network	0.9015	0.9218	0.6926	0.6102	-0.082	Strong

5. Prediction Performance

Table 13 presents full results across all models and splits. All metrics use the original bar scale (exp back-transform). R^2 and RMSE are on log scale for training consistency. Ridge was not submitted — its validation MAPE of 0.41 was not competitive. The Neural Network was submitted based on its best validation MAPE of 0.1321.

Table 13: Full Model Performance — Train, Validation, and Test

Model	Set	R^2	MAPE	RMSE	MAE (bar)	Notes
Ridge Regression	Train	0.7653	0.4524	0.4787	0.1459	Underfitting
Ridge Regression	Val	0.7832	0.4106	0.4663	0.1574	Train $\approx$ Val
Random Forest	Train	0.9939	0.0590	0.0774	0.0226	Mild overfit
Random Forest	Val	0.9572	0.1617	0.2072	0.0659	Strong generalisation
Neural Network (MLP)	Train	0.9903	0.0746	0.0972	0.0240	Mild overfit
Neural Network (MLP)	Val	0.9712	0.1321	0.1699	0.0475	Best validation
Neural Network (MLP)	Test	—	0.1926	—	—	Kaggle private LB ✓

Ridge showed train $\approx$ val R^2 (~ 0.77), confirming underfitting. Random Forest showed mild overfitting (train R^2 0.99 vs val 0.96) expected for fully-grown trees, mitigated by ensemble averaging. The Neural Network achieved the best generalisation with val R^2 0.9712 and val MAPE 0.1321. The Kaggle private leaderboard MAPE of 0.1926 confirms strong test generalisation, achieving 20/20 marks. The gap between val MAPE (0.1321) and test MAPE (0.1926) is expected — the validation set was drawn from the same distribution as training, producing a slightly optimistic estimate. Table 14 maps the Kaggle score to the marking guide.

Table 14: Kaggle Score vs Marking Guide

MAPE Range	Marks Available	This Submission	Marks Achieved
< 0.20	20 / 20	0.1926 ✓	20 / 20
0.20 – 0.23	18 / 20	—	—
0.23 – 0.25	15 / 20	—	—
> 0.50	0 / 20	—	—

6. Self-Reflection

6.1 Challenges

The data cleaning phase was the most demanding aspect. Multiple overlapping issues — missing values, corruption, duplicates, and label inconsistencies — required careful sequencing. For example, imputing before removing duplicates would have preserved incorrect values in copied rows. Distinguishing genuine outliers from legitimate extreme blast pressures also required physical reasoning rather than purely statistical rules; the IQR method flagged 881 target values that were in fact critical training examples.

6.2 Key Insights

The most impactful insight was the log-transformation of the target variable, which improved cross-validated R^2 from ~ 0.72 to ~ 0.90 — a larger gain than any subsequent tuning step. The train-only fitting rule for scalers and imputers was another critical lesson: fitting on the full dataset introduces subtle data leakage with no obvious error. Physics-informed feature engineering also proved more effective than expected, with Tank Volume and Sensor Distance to BLEVE becoming top-ranked features after construction.

6.3 Future Improvements

The most promising extension would be model ensembling. An attempt was made combining Random Forest (30%) and Neural Network (70%), which improved local validation MAPE from 0.1321 to

0.1285. However, the ensemble scored 0.2297 on the Kaggle private leaderboard — worse than the single MLP. This result highlights that local validation improvements do not always transfer to the true test set, particularly when the validation set is not fully independent. Future work would use Bayesian optimisation for more efficient hyperparameter search and establish a pre-committed holdout set before any preprocessing decisions to obtain an unbiased generalisation estimate.

6.4 Conclusion

The final Neural Network model achieved a Kaggle private leaderboard MAPE of 0.1926, placing in the highest marks bracket. The project developed practical skills in systematic data cleaning, physics-informed feature engineering, principled model comparison, and disciplined cross-validation that are transferable across machine learning domains.

Generative AI Acknowledgement

In accordance with Curtin University academic integrity policies, the following discloses all use of generative AI tools in this assignment. All code and written content produced with AI assistance has been reviewed, understood, and verified by the student. The student is able to explain all submitted work in full.

Table A1 details where and how generative AI was used throughout this project.

Table A1: Generative AI Usage Declaration

Tool	Where Used	How Used
Claude (Anthropic)	Data preprocessing	Assisted in structuring the cleaning pipeline (missing value strategy, outlier treatment decisions, duplicate identification). All code was reviewed, tested, and understood before inclusion.
Claude (Anthropic)	Feature engineering	Suggested candidate engineered features. Final selection was made by the student based on RF importance scores and physical reasoning about BLEVE blast wave physics.
Claude (Anthropic)	Model development	Assisted with structuring the model comparison framework and hyperparameter search code. All model choices, search ranges, and final selections were made by the student.
Claude (Anthropic)	Report writing	Assisted in drafting report sections. All technical content, results, and interpretations reflect the student's own analysis and understanding of the work performed.

Important: All Kaggle submissions are the student's own work. The final prediction file was generated by running the student's notebook independently. No prediction files, datasets, or model outputs were sourced from other students or online notebooks.

References

The following sources were consulted and are cited throughout this report. References [1] and [2] are from the assignment specification. Reference [3] is the primary library used for all machine learning models. Reference [4] acknowledges the AI tool used.

Table A2: References

#	Reference
[1]	Jingde Li, Qilin Li, Hong Hao, and Ling Li. Prediction of BLEVE blast loading using CFD and artificial neural network. <i>Process Safety and Environmental Protection</i> , 149:711–723, 2021.
[2]	Qilin Li, Yang Wang, Yanda Shao, Ling Li, and Hong Hao. A comparative study on the most effective machine learning model for blast loading prediction: From GBDT to transformer. <i>Engineering Structures</i> , 276:115310, 2023.
[3]	Scikit-learn: Machine Learning in Python. Pedregosa et al., <i>JMLR</i> 12:2825–2830, 2011. https://scikit-learn.org

#	Reference
[4]	Anthropic. Claude (Version claude-sonnet-4-6) [Large language model]. Used for coding assistance and report drafting. https://www.anthropic.com , 2026.